SHRINATH SHAH linkedin.com/in/shrinath-shah-electronics

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CAREER OBJECTIVE:- Application of my knowledge and skills in VLSI, Electronics to the relevant industry, and to increase my knowledge in that field.

Professional Experience Summary
● Current Hands-on in Cadence Virtuoso tool, ADE-spectre, ADExl- spectre, Maestro and designing Analog amplifier circuit on LTSpice. ● Worked on Basic Unix shell and C programming. ● Worked on VLSI Languages VHDL, Verilog, VerilogA, languages for coding. ● Worked on Embedded C, Assembly language of 8085 and 8051. ● Experience in tools of Cadence like Virtuoso; Mentor Graphics tools like Schematic and Layout; Senataurus TCAD software; Xylink ISE; Keil-µVision. ● Knowledge in Matlab, NgSpice;

Employment Details

Organization Name	Designation	Period
Green Semiconductors Pvt Ltd, (Bangalore, India)	Analog Circuit Design-III	4/12/2023 till date
Cientra TechSolution Pvt. Ltd. (Bangalore, India)	Sr. Analog Design Engineer	28/07/2022 to 27/04/2023
Lead IC Pvt Ltd (Bangalore India)	Analog Design Engineer	22/11/2021 to 15/07/2022
Singularity Labs India Pvt. Ltd. (Gandhinagar, Gujarat, India)	Analog Circuit Design Engineer	17/06/2019 to 31/10/2021
Intel Technology India Pvt. Ltd. (Bangalore, India)	Pre-Si-Verification Intern	31/05/2017 to 26/06/2018

Technical Skills

Primary Skills	Hspice, Cadence Virtuoso: ADE, ADEXL, Maestro; Spectre; VerilogA programming
Technology Nodes	GF 22nm, General 45nm, Tsmc 55nm
Secondary skills	Basic use of VerilogA, Verilog; Elementary knowledge of Embedded C programming, Normal C programming; Elementary Unix-Linux commands and GVIM Editor command

Project A	**Client AMD-XILINK** (Green Semiconductor, LTTS)
Role	AMS-Verification,
Summary	• Initially simulation using Synopysys tool ESP command of circuit for verifying RTL code (behavioural model), and schematic (virtuoso based) for circuit (only simulation of pre-existing some circuits and pre existed behavioural-model based codes) • Design of testbench and extraction of netlist in Hspice format for some testbenches of cadence virtuoso. • Then Hspice based coding for dc, transient and simulation of testbench- netlist and applying measurement coding for single PVT and multi PVT of each testbench. • Porting of some circuit blocks from one foundry to another using AMD-XILINK flow.

Project 1	**Opamp for unity feedback (**Cientra TechSolution Pvt. Ltd. — Bangalore, India)
Role	Analog design and monte carlo simulation (In house project)
Description	Opamp was used to buffer analog i/p voltage in range of 0.4 V to 1.4V for 2.5V supply devices
Responsibilities	• Performance Analysis includes DC, Leakage, Stability-Phase margin analysis wrt Vdd -Temperature variations.
Challenges	• To maintain linearity of output for input range 0.4V to 1.4V in max absolute difference of 0.005 mv across PVT corners by initial designing at 0.9 at TT corner 50 degree temperature at 2.5V Vdd • Improving open loop phase margin to 57-63 degree for input range across corners and Keeping PSRR less than -40 db.
Simulation Analysis	• DC sweep of input voltage • DCOP for Vdsat margin, AC, Stability, PSRR analysis at 3 input voltage • Monte carlo sim for dc sweep analysis. • DC Leakage output current measurement at tristate condition.
Tools Used	Cadence Virtuoso, Maestro Spectre
Technology Node	TSMC 55nm

Project 2	**LDO drop out regulator (**Cientra TechSolution Pvt. Ltd. — Bangalore, India)
Role	Analog design (In house project)
Description	Low Drop Out regulator was used to supply power to blocks internal to chip. The Input voltage was 1V & output was 0.66V, max ILoad of 5mA with 1uF load cap.
Specifications	• Input Vdd variation 0.95V to 1.05V. • Output Load current variation 1mA to 5mA • Load capacitance around 1uF
Responsibilities	In design and verification of LDO that had opamp, power-fet in unity feedback. following analysis were required o be done: •Dc analysis •Ac analysis for 5% Vdd variation for Iload 1mA-5mA , to check stability. •Transient analysis for load and line variation
Challenges	• To maintain linearity of output for input range 0.4V to 1.4V in max absolute difference of 0.005 mv across PVT corners by initial designing at 0.9 at TT corner 50 degree temperature at 2.5V Vdd • Improving open loop phase margin to 57-63 degree for input range
Simulation Analysis	• DC, DC sweep wrt Vdd variation and Iload variation • AC analysis to check stability of opamp-power PFET for loads current 10uA-10mA and Vdd 0.95 to 1.05V • AC analysis to check PSRR • Transient analysis to Check Load and Vdd transient changes.
Tools Used	LT_Spice
Technology Node	General 45nm

Project 3	**Schimitt trigger (**Cientra TechSolution Pvt. Ltd. — Bangalore, India) 1 week

Role	Analog design (In house project)
Description	Design and test of Schimit trigger for given Vsph and Vspl using 6 transistors (3 pmos and 3 nmos) and to check effect of input rise time and fall time on Vsph and Vspl for load of 1pF
Responsibilities	In design and test of Schimit trigger, following analysis were required o be done: • Dc analysis at Vin equal to Vdd (1) and 0(V). • DC sweep analysis for input and falling input and to observe output and inside node voltage • Transient analysis of output for input, rise and fall time variation and input pulse width variation for square wave input. • Transient analysis for sinosoidal input at 100KHz for 1pf load.
Simulation Analysis	• Simple DC, DC_Sweep,Transent
Tools Used	LT_Spice
Technology Node	General 45nm

Project 4	**Dgital Inverter design (**Lead IC Pvt. Ltd. — Bangalore, India)
Role	Digital design (In house project)
Description	Design and test of Inverters for given rise time, fall time and max operating frequency and for same design I found maximum dc current and maximum, average transient current at specified frequency from transient analysis
Responsibilities	• Performance Analysis includes DC, Transient Analysis, Leakage analysis wrt Vdd-Temperature variations.
Simulation Analysis	• Simple DC and DC sweep of input voltage (0 to 1v) • Transient analysis at 1Khz square wave to check rise time, fall time, propagation delay, max Vdd current, average Vdd current
Tools Used	Cadence Virtuoso, ADE Spectre
Technology Node	tsmcN28

Project 5	**IM2CAL (**Singularity Labs India Pvt. Ltd. – Gandhinagar, Gujarat, India)
Role	AMS design and verification (In house project)
Description	Im2cal is mixed-signal, calibration block to I & Q channel of mixer-receiver. IM2cal output is connected to gate of nfet of mixer of Receiver and IM2cal output voltage can be used to either remove or add offset between Positive and Negative output of either I-ch Mixer or Q-ch Mixer or both Channel Mixer. It was designed using combination of current steering DAC, Unity gain Analog Buffer and digital resistor selector block for increasing output analog voltage from input or decreasing output voltage from input or keeping output equal to input
Specifications	• Input analog voltage dc variation from 350 mV to 600 mv of low-pass frequency range 0 to 1KHz • Input 9 bit, 0 to 255 input to decrease output voltage dc value from input dc voltage, and 256 to 511 input range to increase output voltage dc value from input dc voltage • Load Mixer gate of gate capacitance around 20 pF Supply Voltage source 0.8V
Responsibilities	• Design & Simulation of the block amp wrt PVT variations • Maintaining unity gain from 0.9 to 1 of closed loop design with temperature changes from -40 degree C to 125 degree C in simulation with or without offset • VerilogA coding wrt results of TT_50 corner.
Challenges	• Keeping off leakage current less than 50 nA. Meeting stringent specifications in Output swing

Simulation Analysis	● Simple DC and Parametric DC ● Transient analysis with Rx (Receiver chain) components- LNA (Low noise Amplifier), Rx-Mixer and TIA (TransImpedance Amplifier) ● Dc, parametric DC and Transient sim for Extracted R, Extracted RC of layout view and explaining layout team regarding any changes in layout for less effect of parasitics.
Tools Used	Cadence Virtuoso, ADE Spectre
Technology Node	GF cmos22fdsoi

Project 6	**Current output DAC (**Singularity Labs India Pvt. Ltd. – Gandhinagar, Gujarat, India)
Role	AMS design and verification(In house project)
Description	8 bit DAC using topology of combination of "Current Steering DAC using binary- weighted current sources" and "Generic current steering DAC", for obtaining constant different magnitude currents in specified range for given almost constant load and for multi-purpose block operation. Minimum on current 100nA and max on current 25uA, step-size 100nA. Load 400mV load of low-pass frequency range 0 to 100KHz
Responsibilities	● Maintaining Step-Size variation less than 10% with temperature changes and keeping minimum on current variation 10% with temperature changes from -40 degree C to 125 degree C in simulation ● Maintaining Linearity wrt PVT corners ● Simulations with 5% supply (Vdd 0.8V) variations VerilogA coding wrt results of TT_50 corner.
Challenges	● Keeping, off (leakage) current of block at output less than 10nA at temperature 50 degree C for TT corner. Maintaining max on current for no more than 10% variation at FF corner
Simulation Analysis	● Simple DC and Parametric DC ● Safe operating area (SOA) reliability analysis of block and inside sub-block and transistor for reliability check under off condition at DC using Cadence. ● Dc, parametric DC and sim for Extracted R, Extracted C of layout view and explaining layout team regarding any changes in layout for less effect of parasitics.
Tools Used	Cadence Virtuoso, ADE and ADEXl Spectre
Technology Node	GF cmos22fdsoi

Project 7	**One Stage Opamp single ended output (**Singularity Labs India Pvt. Ltd. – Gandhinagar, Gujarat, India)
Role	Analog design and verification(In house project) 3 months
Description	Design of one-stage op-amp for unity feedback application, I.e Dc analog buffer for 20pF load, supply (vdd) 0.8V and frequency 10 to 500KHz Circuit topology single stage "open loop high gain differential Amplifier with active current mirror and n-mosfet as tail current source
Responsibilities	● Design & Simulation of this Op-amp wrt PVT variations ● Analysis of gain, Phase margin and closed loop output voltage for Input range 0.4V to 0.6V. ● Open loop should be stable ● Maintaining gain from 0.9 to 1 of closed loop design with temperature changes from -40 degree C to 125 degree C in simulation for PV variations. VerilogA coding wrt results of TT_50 corner.
Challenges	● Keeping off leakage current less than 20 nA.

	• Maintaining Phase margin above 60 degree of open loop design across PVT and given Input range.
Simulation Analysis	• Simple DC and Parametric DC across input range 0.4 to 0.6V • AC analysis to check open loop gain and stability • Transient analysis in closed loop • Safe operating area (SOA) reliability analysis of block and inside sub-block and transistor for reliability check under off condition at DC using Cadence. • Dc, parametric DC and Transient sim, open loop AC for Extracted R, Extracted RC of layout view and explaining layout team regarding any changes in layout for less effect of parasitics.
Tools Used	Cadence Virtuoso, ADE Spectre
Technology Node	GF cmos22fdsoi

Project 8	**Logic gates (**Singularity Labs India Pvt. Ltd. – Gandhinagar, Gujarat, India)
Role	Digital logic design and verification (In house project)
Description	Initially I designed and tested Inverters for given rise time, fall time and max operating frequency and for same design I found maximum dc current and maximum, average transient current at specified frequency from transient analysis
Responsibilities	• Performance Analysis includes DC, Transient Analysis, Leakage analysis wrt PVT variations. • Simulation with temperature changes from -40 degree C to 125 degree C and 5 percent supply (Vdd 0.8V) variations • VerilogA coding wrt results of TT_50 corner.
Challenges	• Keeping dc current less than 10nA with minimum area across PVT corner.
Simulation Analysis	• Simple DC and DC sweep of input voltage • Transient analysis at 10Mhz square wave to check rise time, fall time, propagation delay, max Vdd current, average Vdd current • Safe operating area (SOA) reliability analysis of block and inside transistor for reliability check under off condition at DC using Cadence
Tools Used	Cadence Virtuoso, ADE Spectre
Technology Node	GF cmos22fdsoi

Project 9	**Pre-Si verification intern (**Intel Technology India Pvt Ltd – Bangalore, India)
Role	Pre Silicon verification (In house project)
Summary	• Here I worked on regression for different test cases. • Learnt, worked, and executed Perl scripting for different purposes. • Worked on writing, running, and debugging test cases on different blocks of Intel project using system Verilog and debugging using VCS tool, Verdi, DVE wave form analyser. • Provided effective communication on status of the progress, problem and solution to team. I learnt and worked on UVM for MTech dissertation work

Other Technical Skills using Cadence Virtuoso: Elementary knowledge of Monte Carlo simulation thorough ADEXL. Worked on ‘stb’, ‘sp’, ‘noise’, ‘pac’ analysis using ADEL spectre.

Teaching Assistant in SVNIT Mtech :- Worked as a Teaching Assistant during my college tenure, supporting students. July 2016 to December 2016, handled undergraduate first year basic electronics laboratory and supervision of internal exam and practical laboratory exam. January 2017- May 2017 handled undergraduate third year Digital Integrated Circuit laboratory and supervision of internal exam and practical laboratory exam. Handled student paperwork and supported faculty with classroom and administrative tasks

Education:

Year	Degree	Major Subject	Institution	CGPA/ Percentage	Full time/Part time
2016-2019	M.TECH	VLSI & Embedded Systems	S.V.N.I.T, Surat Gujarat	7.75 C.G.P.A	Full time
2012-2016	B.E	Electronics	MSU Baroda, Gujarat, India	8.58 C.G.P.A	Full time
2012	H.S.C (PUC, 12 Sci Eng medium)	PCM	G.H.S.E.B (Gujarat)	85 Percent	Full time
2010	S.S.C (Pre-Puc, 10 General, Eng medium)		G.S.E.B (Gujarat)	82 Percent	Full time

PROJECTS

M. Tech:- Major Dissertation:-

Performing Pre-Si Verification of Register based ALU using UVM.:- It requires thorough knowledge at black box level of IP, (here) register based ALU, which is to be verified. Then using System Verilog and UVM, test bench is to be developed for IP to be verified. Then finding major errors, bugs by running test cases and to find functional coverage of required operations by running regression and to reach error free results.

M. Tech:- Academic Projects:-

1. 8x1 Multiplexer using 180nm technology and sizing using logical effort in mentor graphics schematic and layout.
2. Fabrication and characterization of 1N4007 PN junction diode using Senatrus TCAD and realized its characteristics using SDEVICE.
3. Design and implementation of differential amplifier in mentor graphics.

B.E:-Final year major dissertation on “RFID card-based bike security and GSM based accident informing system”.

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ACHIEVEMENTS

Secured a GATE(ECE) score of 648 and AIR 1495 in 2016
Winner of Bournvita Quiz at school level.
Always in Top 3 in school.

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TECHNICAL-COURSES

1) “SOC Verification using System Verilog” online course by Udemy (2019)

2) “Learn System Verilog Assertions and Coverage Coding” online course by Udemy (2019)
3) “German-I” online certificate course by NPTEL (Sep-Dec 2020) overall score 63%.
4) “Body Language: Key to Professional Success” online certification course by NPTEL (Sep-Oct 2020) overall score 82%
5) “Effective Writing” online certification course by NPTEL (Jan-Feb 2021) overall score 73%

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ADDITIONAL INFORMATION

HOBBIES & INTERESTS:

Travelling, Improving technical skills, reading books, Light-Music songs listening.

PERSONAL PROFILE:

Languages known: English, Gujarati, Hindi, German-1 (elementary)

Year of birth: 1994

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Date:- May 07, 2026; Place :- India.